

CST8809_010 Business Continuity & Disaster Recovery

Main Assignment 2

Group - 6

Professor- Robson Minjoro

Members Names

Nafisa Osman

Oderinlo Oluwatoni

Darshil Patel

Patrick Edeh

Prathamesh Ghosh

Shaikshahils Mohammedsheriff

Date: 08-Mar-2024

Table of Contents

1. OVERVIEW.....	3
1.1 PURPOSE	3
2. ASSUMPTION	4
3. SYSTEM DESCRIPTION.....	4
4. BIA DATA COLLECTION.....	5
4.1 DETERMINE PROCESS AND SYSTEM CRITICALITY	5
4.2 STAKEHOLDERS.....	6
4.2.1 CALL TREE	7
4.3 BUSINESS PROCESSES.....	8
4.3.1 IDENTIFY OUTAGE IMPACTS AND ESTIMATED DOWNTIME	9
4.3.2 OUTAGE IMPACT.....	10
4.3.3 ESTIMATED DOWNTIME	10
4.4 IDENTIFY RESOURCE REQUIREMENTS	12
4.5 IDENTIFY RECOVERY PRIORITIES FOR SYSTEM RESOURCES.....	12
4.6 RECOVERY PLAN.....	13
4.7 TEST PLAN.....	14
5 RISK ANALYSIS	16
5.1 RISK COST ANALYSIS	16
5.2 RISK RANKING	19
5.3 OVERALL SYNTHESIS OF MATERIAL	19
6 RECOMMENDATIONS	20
7 REFERENCES	23

1. OVERVIEW

The Business Impact Analysis BIA is developed as part of the contingency planning process for the CanTech data center. This analysis will help us understand the critical functions, dependencies, and recovery priorities necessary to minimize the impact of disruptions and maintain business continuity. It was prepared on March 08, 2024.

1.1 PURPOSE

The purpose of the Business Impact Analysis (BIA) for CanTech Datacenters is to assess the potential impacts of disruptions to the organization's operations and services. It aims to identify and prioritize critical functions and dependencies, correlating them to the mission/business processes supported by the data center infrastructure. By characterizing the impact of system disruptions on these processes, the BIA helps determine recovery priorities and resource requirements necessary for maintaining business continuity.

The BIA consists of the following steps:

- **Determine mission/business processes and recovery criticality:** Identify the mission/business processes supported by the data center infrastructure and assess the impact of system disruptions on these processes. Evaluate outage impacts and estimate downtime, considering the maximum tolerable downtime for maintaining business operations.
- **Identify resource requirements:** Conduct a comprehensive evaluation of the resources needed to resume mission/business processes and related interdependencies swiftly. This includes identifying facilities, personnel, equipment, software, data files, system components, and vital records required for recovery efforts.
- **Identify recovery priorities for system resources:** Based on the results of the previous steps, establish priority levels for sequencing recovery activities and allocating resources effectively. Link system resources to critical mission/business processes to determine recovery priorities accurately.

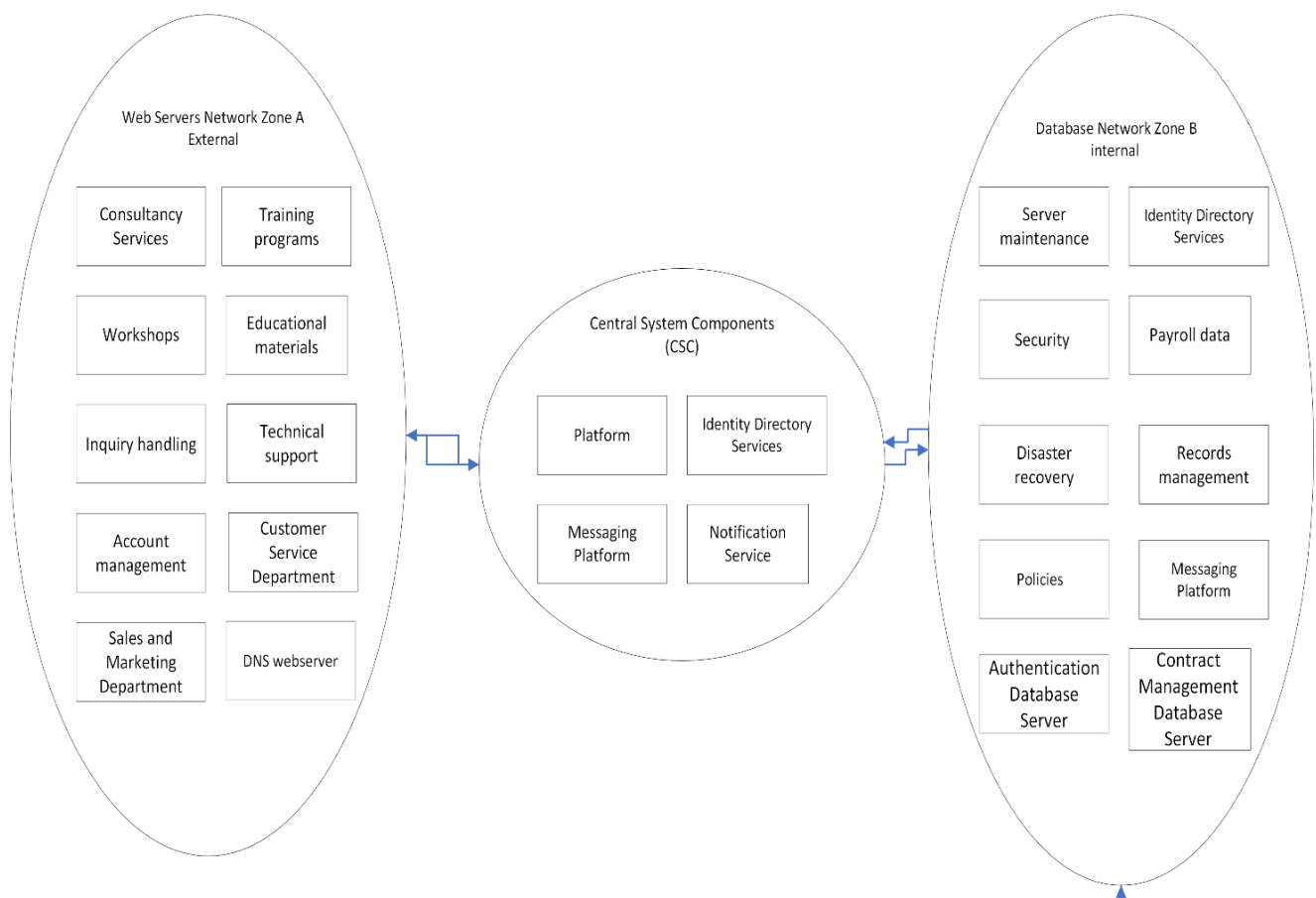
This BIA document serves as a foundation for building the CanTech Datacenters' Information System Contingency Plan (ISCP) and is an integral component of the ISCP framework.

2. ASSUMPTION

During the development of this Business Impact Analysis (BIA) for CanTech Datacenters, the following assumptions were made:

- The information provided in the BIA is based on the current operational state of CanTech Datacenters and its facilities.
- It is assumed that the identified critical functions and dependencies accurately represent the core operations within CanTech Datacenters.
- The BIA assumes the availability of necessary resources, including personnel, technology, and financial support, to execute the proposed recovery plans effectively.

3. SYSTEM DESCRIPTION



The diagram is divided into three main zones:

1. **Web Servers Network Zone A (External):** This area includes services provided to external clients such as consultancy services, training programs, workshops, educational materials, inquiry handling, technical support, account management, and the departments responsible for these services like Sales and Marketing, Customer Service, etc.
2. **Central System Components (CSC):** This zone appears to be the core where platform services, messaging, identity directory services, and notification services are managed. This could represent the central operational systems that facilitate both external-facing services and internal processes.
3. **Database Network Zone B (Internal):** This zone seems to focus on internal functions such as server maintenance, identity directory services, security, disaster recovery, records management, policies, authentication, contract management, and the handling of sensitive data like payroll.

4. BIA DATA COLLECTION

4.1 DETERMINE PROCESS AND SYSTEM CRITICALITY

CanTech's ecosystem comprises a broad spectrum of stakeholders, ranging from the executive team, including the CEO, COO, CFO, and the Infrastructure Manager, to specialized roles such as IT and Database Admins, Network and Systems Admins, and Customer Service Executives. This diverse team underscores the company's reliance on a variety of functions and processes that support its mission and business operations.

Business Process	Department	Internal/External
Server maintenance	IT and Infrastructure Department	External
Data recovery	IT and Infrastructure Department	External
Data recovery	IT and Infrastructure Department	External
Consultancy Services	Consultancy Department	External
Strategic planning	Consultancy Department	External
Assessments	Consultancy Department	External
Recommendations	Consultancy Department	External
Training programs	Training and Development Department	External

Workshops	Training and Development Department	External
Educational materials	Training and Development Department	External
Network monitoring	Network Operations Department	Internal
Security	Network Operations Department	Internal
Upgrades	Network Operations Department	Internal
Software maintenance	IT Department	Internal
IT support	IT Department	Internal
Compliance	IT Department	Internal
Inquiry handling	Customer Service Department	External
Technical support	Customer Service Department	External
Account management	Customer Service Department	External
Physical security	Security and Infrastructure Department	Internal
Risk assessments	Security and Infrastructure Department	Internal
Disaster recovery	Security and Infrastructure Department	Internal
Client targeting	Sales and Marketing Department	External
Marketing campaigns	Sales and Marketing Department	External
Market research	Sales and Marketing Department	External
Budgeting	Finance and Accounting Department	Internal
Financial reporting	Finance and Accounting Department	Internal
Payroll	Finance and Accounting Department	Internal
Onboarding	Human Resources Department	Internal
Records management	Human Resources Department	Internal
HR policies	Human Resources Department	Internal

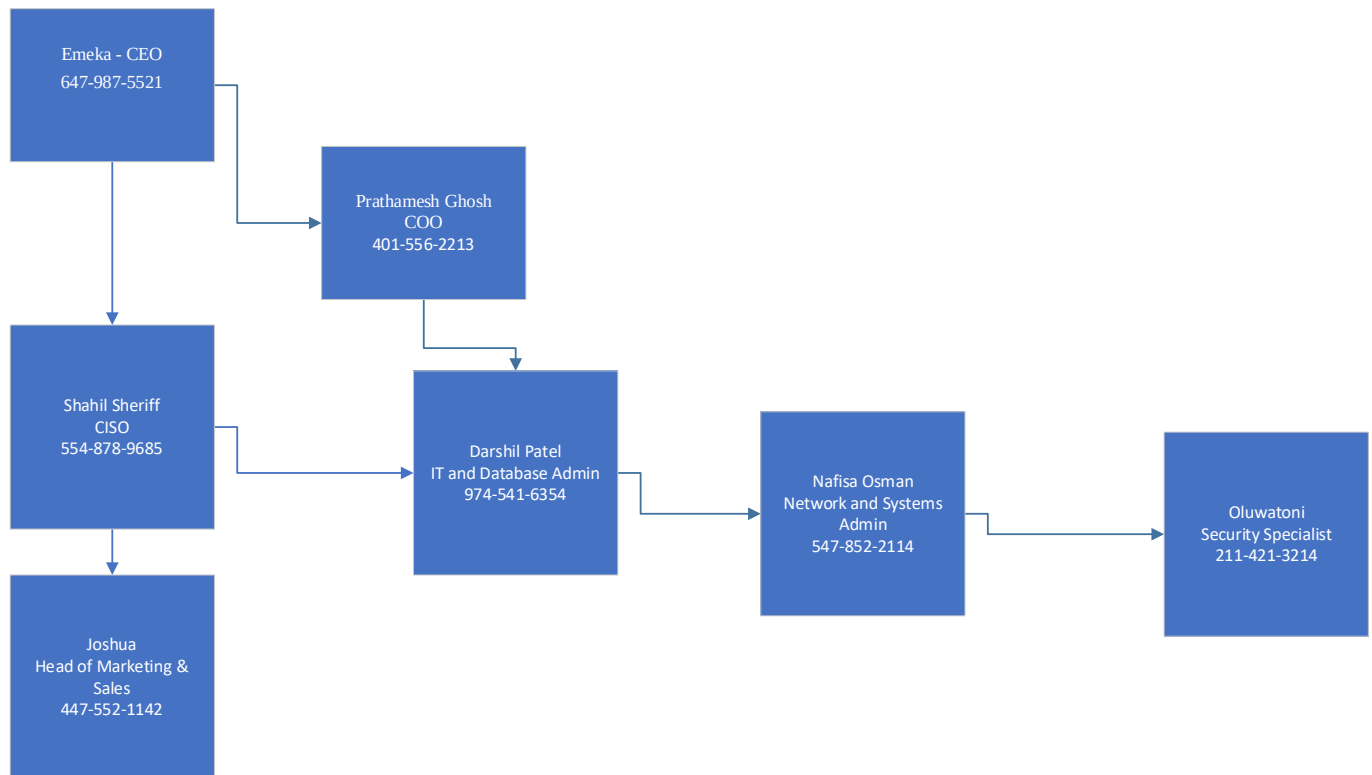
4.2 STAKEHOLDERS

The prioritising of key stakeholders at CanTech, underscores a strategic focus on leadership and core technical functions critical to the organization's operations. The Chief Executive Officer, Infrastructure Manager, IT and Database Admin, Network and Systems Admin, and Security Specialist are identified as essential, with priorities from one to five, reflecting their roles in decision-making, infrastructure integrity, data management, network security, and overall system protection.

No.	Stakeholders	Key Stakeholder (X)	Priority Order
1	Chief Executive Officer	X	1
2	Infrastructure Manager	X	2
3	IT and Database Admin	X	3
4	Network and Systems Admin	X	4
5	Security Specialist	X	5
6	Server and Storage Admin		
7	Electrical and Mechanical Engineer		
8	Human Resources and Admin Manager		
9	Marketing and Sales Executives		
10	Customers and Clients		

4.2.1 CALL TREE

Designing a call tree involves structuring the communication flow starting from the top management down to all relevant staff, ensuring everyone is informed and knows their responsibilities in case of specific incidents, such as data center outages, security breaches, or natural disasters.



4.3 BUSINESS PROCESSES

CanTech Data Center offers a broad spectrum of services crucial for both corporate and individual clients' operational success. Below are the key business processes and their descriptions along with its criticality.

Business Process	Department	Description	Internal / External	Criticality
Server maintenance	IT and Infrastructure Department	Upkeep of core servers	External	High
Data recovery	IT and Infrastructure Department	Restoring lost data	External	High
Data recovery	IT and Infrastructure Department	Restoring lost data	External	High
Consultancy Services	Consultancy Department	Expert advice provision	External	Medium
Strategic planning	Consultancy Department	Long-term planning aid	External	Medium
Assessments	Consultancy Department	Evaluating IT systems	External	Medium
Training programs	Training and Development Department	Skills development	External	Low
Workshops	Training and Development Department	Interactive learning	External	Low
Network monitoring	Network Operations Department	Supervising network flow	Internal	High
Security	Network Operations Department	Safeguarding data	Internal	High
Upgrades	Network Operations Department	Systems improvement	Internal	Medium
Software maintenance	IT Department	Software updates	Internal	High
IT support	IT Department	Technical assistance	Internal	High
Compliance	IT Department	Regulatory adherence	Internal	High
Inquiry handling	Customer Service Department	Responding to queries	External	High
Technical support	Customer Service Department	Technical issue solving	External	High
Account management	Customer Service Department	Client account oversight	External	Medium
Physical security	Security and Infrastructure Department	Facility protection	Internal	High

Risk assessments	Security and Infrastructure Department	Risk analysis	Internal	High
Disaster recovery	Security and Infrastructure Department	Emergency response	Internal	High
Client targeting	Sales and Marketing Department	Attracting new clients	External	Medium
Marketing campaigns	Sales and Marketing Department	Promoting services	External	Medium
Market research	Sales and Marketing Department	Market analysis	External	Low
Budgeting	Finance and Accounting Department	Financial planning	Internal	Medium
Financial reporting	Finance and Accounting Department	Reporting finances	Internal	Medium
Payroll	Finance and Accounting Department	Employee payment	Internal	High
Onboarding	Human Resources Department	New hire integration	Internal	Medium
Records management	Human Resources Department	Documentation upkeep	Internal	Medium
HR policies	Human Resources Department	Policy enforcement	Internal	Medium

4.3.1 IDENTIFY OUTAGE IMPACTS AND ESTIMATED DOWNTIME

The below section identifies and characterizes the types of impact categories that a system disruption is likely to create.

Business Process	Impact	Financial Loss	Reputation	Regulatory Compliance	Client Service Delivery	Data Integrity
Server Maintenance	High	Severe	Severe	Moderate	Severe	Severe
Data Recovery	High	Severe	Severe	Moderate	High	Severe
Network Monitoring	High	Moderate	High	Moderate	High	High
Software Maintenance	High	Moderate	High	Moderate	High	High
IT Support	High	Moderate	Moderate	Moderate	Severe	Moderate
Inquiry Handling	High	Low	Severe	Low	Severe	Low
Account Management	High	Moderate	Severe	Low	Severe	Moderate

Risk Assessments	High	Low	Moderate	Severe	Moderate	High
Client Targeting	Medium	Low	Moderate	Low	Moderate	Low
Marketing Campaigns	Medium	Low	High	Low	High	Low
Financial Reporting	High	Severe	High	Severe	Moderate	High
Payroll	High	Severe	High	Severe	Severe	Moderate

4.3.2 OUTAGE IMPACT

The analysis of process and system criticality demonstrates the essential nature of CanTech's services and the significant impact a system disruption could have on its operations and clients.

- Financial Loss: CanTech faces potential financial losses in terms of revenue and compensations to clients. Service Level Agreements (SLAs) with guaranteed uptime could result in penalties.
- Reputation: The reliability of CanTech as a data center provider is crucial. Downtime affects client trust and can lead to a loss of current and future business.
- Operational: Delays in internal and client-related projects, disruption to internal operations, and additional overtime work to recover from the outage.
- Estimated Downtime Tolerance: Given CanTech's critical role in hosting and consultancy services, the maximum tolerable downtime (MTD) is estimated at 24 hours for hosting services due to the high dependency of clients on these services for their business operations. For consultancy and training services, a slightly longer MTD of 48 hours may be tolerable, considering the possibility of rescheduling.

This highlights the need for a robust disaster recovery plan to minimize downtime and mitigate the risks identified.

4.3.3 ESTIMATED DOWNTIME

Working directly with mission/business process owners, departmental staff, managers, and other stakeholders, estimate the downtime factors for consideration as a result of a disruptive event.

- **Maximum Tolerable Downtime (MTD).** The MTD represents the total amount of time leaders/managers are willing to accept for a mission/business process outage or disruption and includes all impact considerations.

Determining MTD is important because it could leave continuity planners with imprecise direction on (1) selection of an appropriate recovery method, and (2) the depth of detail which will be required when developing recovery procedures, including their scope and content

- **Recovery Time Objective (RTO).** RTO defines the maximum amount of time that a system resource can remain unavailable before there is an unacceptable impact on other system resources, supported mission/business processes, and the MTD. Determining the information system resource RTO is important for selecting appropriate technologies that are best suited for meeting the MTD.
- **Recovery Point Objective (RPO).** The RPO represents the point in time, prior to a disruption on system outage, to which mission/business process data must be recovered - given the most recent backup copy of the data- after an outage.

The table below identifies the MTD, RTO, and RPO for the organizational mission/business processes that rely on the following table

Business Process	MTD	RTO	RPO
Server Maintenance	4 hours	2 hours	1 hour
Data Recovery	4 hours	1 hour	30 minutes
Network Monitoring	2 hours	1 hour	15 minutes
Software Maintenance	4 hours	2 hours	1 hour
IT Support	4 hours	2 hours	1 hour
Inquiry Handling	4 hours	2 hours	1 hour
Account Management	24 hours	8 hours	4 hours
Risk Assessments	4 hours	2 hours	1 hour
Client Targeting	8 hours	4 hours	2 hours
Marketing Campaigns	24 hours	12 hours	6 hours
Financial Reporting	24 hours	8 hours	4 hours
Payroll	48 hours	24 hours	12 hours

4.4 IDENTIFY RESOURCE REQUIREMENTS

The following table identifies the resources including hardware, software and other resources such as data files.

System Resource / Component	Platform/OS/Version	Description
Core Database Server	SQL Server on Windows Server	Critical data storage
Primary Web Server	Apache on Linux	Client-facing web services
Network Infrastructure	Cisco IOS or NX-OS	Connectivity and security
Email Server	Microsoft Exchange Server	Internal and external communication
Customer Management System	Salesforce on Cloud	Client relationship management
Backup Server	Windows Server with Backup Exec	Data backup and recovery
Development Server	Ubuntu with Docker	Development and testing
Office Workstations	Windows 10 Professional	Daily business operations
Security System	Proprietary Security OS	Physical and information security
HVAC Control System	Embedded OS	Environmental control

4.5 IDENTIFY RECOVERY PRIORITIES FOR SYSTEM RESOURCES

In order to identify recovery priorities for system resources for the CanTech scenario, we would analyze the business impact of each system component being down and then prioritize their recovery based on the RTO that aligns with the MTD for the business processes they support.

Priority	System Resource/Component	Recovery Time Objective (RTO)
1	Core Database Servers	2 hours to restore from backups
2	Primary Web Servers	4 hours to rebuild or replace
3	Network Infrastructure	4 hours to restore critical switches
4	Customer Management Systems	8 hours to restore service
5	Email Servers	12 hours to restore from backups
6	Internal Communication Systems	24 hours to replace hardware
7	Backup Servers	24 hours to restore services
8	Development Servers	48 hours to rebuild or replace
9	Non-Essential Support Systems	72 hours to restore or replace

4.6 RECOVERY PLAN

The objective of the CanTech Recovery Plan is to restore critical operations in the event of a system disruption or disaster to minimize impact on operations, maintain customer trust, and comply with regulatory requirements.

Important Recovery Steps:

1. Immediate Response:

- Assess the extent of the disruption.
- Communicate with the emergency response team and stakeholders.
- Secure the site if the disruption involves physical damage or security breach.

2. Resource Recovery:

- Prioritize the recovery of resources based on the criticality, as defined in the recovery priorities table.
- Follow the Recovery Time Objectives (RTO) to restore each system resource.
- Use backups to restore data integrity and service continuity.

3. Systems Recovery:

- Begin with restoring core database servers to ensure data integrity.
- Proceed to network infrastructure to re-establish connectivity.
- Recover customer management systems to resume client services.

4. Testing and Validation:

- Ensure that restored systems are functioning as expected.
- Test the ability to perform business processes against the restored systems.
- Validate with end-users and stakeholders that services are operational.

5. Communications:

- Update customers, employees, and stakeholders on the status of recovery efforts.
- Communicate the restoration of services and any temporary changes in operations.

6. Monitoring and Support:

- Monitor systems for stability and performance issues post-recovery.
- Provide support to address any issues arising from the recovery process.

4.7 TEST PLAN

Purpose:

To validate the effectiveness of the CanTech Recovery Plan through scheduled tests and drills, ensuring that recovery procedures are functional, efficient, and up to date.

Scope:

This plan encompasses testing of all critical systems and processes identified in the Recovery Plan, including IT infrastructure, applications, data recovery procedures, and business continuity measures.

Assumptions:

- All systems are operational and in a known good state at the time of testing.
- Necessary resources, including personnel and equipment, are available for testing.
- Tests will not disrupt normal business operations.

Test Schedule:

- **Frequency:** The recovery processes will be tested semi-annually, with additional tests following significant system changes.
- **Timing:** Tests will be scheduled during off-peak hours to minimize potential disruptions.

Test Types:

1. **Tabletop Exercises:** Simulated discussions conducted in a meeting format to walk through the recovery steps.
2. **Walk-through Drills:** Team members perform their tasks as if a real disaster occurred, without restoring systems.

3. **Technical Recovery Tests:** Actual restoration of systems from backups to an alternate location.
4. **Full Interruption Tests:** Complete shutdown and recovery of a system to validate RTO and RPO.

Test Scenarios:

- Scenario 1: Server failure and data loss.
- Scenario 2: Network outage.
- Scenario 3: Full data center downtime.

Roles and Responsibilities:

- **Test Manager:** Oversees the test plan and ensures completion of objectives.
- **Technical Teams:** Execute the recovery of their respective systems and processes.
- **Observers:** Monitor and document the test process, noting any issues or deviations.

Success Criteria:

- Recovery steps are completed within the defined RTO and RPO for each system.
- Systems are restored to a fully operational state.
- Participants demonstrate familiarity with their roles and responsibilities.

Documentation:

- Develop detailed test scripts that outline steps for execution.
- Record issues, anomalies, and deviations from the plan.

Reporting:

- Prepare a test report summarizing the findings, including any weaknesses or gaps identified.
- Report to be reviewed by the Business Continuity Management Team.

Review and Update:

- Conduct a post-test review session to discuss the outcomes and lessons learned.
- Update the Recovery Plan and Test Plan based on test results and feedback.

5 RISK ANALYSIS**5.1 RISK COST ANALYSIS**

The table below shows the original risk costs, the mitigation costs, and the new costs with the justification on the importance of mitigating the risk.

Risk	Original Risk Cost	Mitigation cost	New Cost	Justification	Risk Level (From most expensive)
Cyber Attacks	\$12.12 million	\$5 million	\$7.12 million	The decrease in expenses related to cyber attacks, from \$12.12 million to \$7.12 million, can be attributed to a \$5 million allocation towards cybersecurity initiatives. These measures likely encompass enhancements in network security, implementation of sophisticated intrusion detection systems, regular software updates, and educational programs for employees. Taking proactive steps to address vulnerabilities strengthens defenses, thereby diminishing the likelihood of successful attacks and minimizing financial repercussions stemming from data breaches, system downtimes, and regulatory penalties. This underscores the effectiveness of substantial investments in cybersecurity to mitigate risks and bolster the organization's resilience against evolving threats. By upholding its reputation and fostering trust among customers, these measures not only cut costs but also elevate the overall	3

				cybersecurity stance, ensuring continuous protection in an ever-changing threat landscape.	
Natural Disasters	\$21 million	\$10 million	\$11 million	The decrease in the cost of the risk of natural disasters from \$21 million to \$11 million can be attributed to a mitigation strategy that saved \$10 million. Likely, this strategy involved implementing measures such as reinforced infrastructure, disaster-resistant building materials, early warning systems, and contingency plans. By investing in these mitigations, the organization has significantly reduced the potential financial impact of natural disasters. This demonstrates the effectiveness of proactive risk management in minimizing losses and ensuring business continuity. Overall, the reduced cost underscores the value of strategic investments in mitigating the impacts of natural disasters on the organization.	1
Global Health Issues	\$250,000	\$100,000	\$150,000	The reduction in the cost of the risk of global health issues from \$250,000 to \$150,000 can be attributed to mitigation efforts that saved \$100,000. These efforts may include implementing health and safety protocols, providing vaccinations, and investing in medical infrastructure. By proactively addressing health risks, the organization has minimized potential financial losses, highlighting the effectiveness of preventive measures in reducing overall costs.	5
Regulatory Changes	\$4 million	\$2 million	\$2 million	The reduction in the cost of regulatory changes from \$4 million to \$2 million reflects savings of \$2 million due to mitigation strategies. Likely, these strategies involve thorough compliance assessments,	4

				proactive adjustments to policies and procedures, and investments in regulatory technology. By implementing these measures, the organization has streamlined its regulatory compliance processes, minimizing potential penalties and operational disruptions. This highlights the value of proactive risk management in reducing overall costs associated with regulatory compliance.	
Technology Advancements	\$18 million	\$9 million	\$9 million	The decrease in costs associated with technology advancements, from \$18 million to \$9 million, reflects savings of \$9 million due to successful mitigation strategies. These strategies likely entail investments in research and development, technology upgrades, and workforce training to embrace emerging technologies. By staying abreast of technological shifts, the organization has curtailed expenses related to outdated systems, inefficiencies, and potential disruptions. This highlights the significance of proactive innovation and strategic technology investments in minimizing risks and optimizing operational costs, ensuring competitiveness and sustainability amidst rapid technological evolution.	2

5.2 RISK RANKING

Ranking:

- Natural Disasters: This risk holds the top position due to its potential for widespread devastation. Events like earthquakes, hurricanes, and floods can inflict significant damage on infrastructure and lives, resulting in immense recovery costs.
- Technology Advancements: While offering benefits, rapid technological changes can introduce risks like system failures and cybersecurity threats. Continuous investment in research, development, and upgrades is crucial to stay competitive and avoid operational setbacks.
- Cyber Attacks: Increasing reliance on digital systems makes organizations vulnerable to data breaches and financial losses. Effective cybersecurity measures, including network security and employee training, are essential to mitigate these risks.
- Regulatory Changes: Compliance with evolving regulations incurs costs such as legal fees and operational adjustments. Maintaining compliance requires ongoing monitoring and implementation of appropriate policies and procedures.
- Global Health Issues: While impactful on society, the direct financial costs of health crises may be relatively lower. Measures such as health protocols and healthcare benefits help minimize disruptions to business operations.

5.3 OVERALL SYNTHESIS OF MATERIAL

Overall, from the risk analysis the significance of implementing risk mitigation strategies to diminish potential financial burdens across diverse risk categories. Cyber attacks, natural disasters, regulatory changes, global health issues, and technology advancements each present distinct challenges and associated costs for organizations. Mitigation measures such as cybersecurity initiatives, infrastructure reinforcement, compliance assessments, health protocols, and technology upgrades have demonstrated effectiveness in lowering these risks and minimizing related expenses. The risk ranking highlights the varying degrees of financial impact and complexity in mitigation efforts across different risk categories, with natural disasters posing the highest risk due to their potential for extensive damage. Ultimately, proactive risk management and strategic investments are vital in protecting organizations against evolving threats and ensuring uninterrupted business operations.

6 RECOMMENDATIONS

- 1. Enhancement of Data Backup and Recovery Systems:** Since data is essential in the maintenance of business continuity, CanTech should make robust investment in different backup and recovery solutions. Some of the backup and recovery solutions they should consider include but not limited to the following:
 - constant backups of all critical data in all their operational centers,
 - the deployment of off-site storage solutions to protect against any form of local disasters, and
 - the use of real-time data reproduction technologies to reduce the loss of data.
- 2. Implementation of Redundant Systems and Network Infrastructure:** CanTech should consider the implementation of redundant systems and network infrastructure so that there can be a reduction in the risk of system downtime. Examples of these considerations should comprise the following:
 - the presence of failover facilities for critical systems,
 - redundant network paths in order to prevent or reduce to the barest minimum single points of failure, and
 - the deployment of high-availability configurations for critical infrastructure sections.
- 3. Development of a Thorough Disaster Recovery Plan (DRP):** Despite the presence of BIA, there should be a detailed and implementable disaster recovery plan. The structure of the plan should entail the following:
 - the steps and processes to be taken should any form of disaster scenario occurs,
 - the assignment of responsibilities to team members, and
 - inclusion of contact information for all key personnel.
- 4. Conducting Frequent Risk Assessments and BIA Updates:** CanTech should carry out regular risk assessments and update its BIA. This is because the risk landscape and business operations are continually changing. The proposed regular risk assessments and update should include the following:
 - re-examining the severity or otherwise of business processes,
 - evaluating new technologies and their impact, and
 - taking into consideration new developments within the regulatory environment.
- 5. Improving the Physical and Cyber Security Measures:** There should be ongoing enhancement in CanTech's security measures. This is to help to protect against both physical and cyber threats. These security measures include, but not limited to the following:
 - implementation of robust access controls,
 - surveillance and monitoring systems for physical security, and
 - utilizing sophisticated cybersecurity measures like firewalls and intrusion detection/prevention systems.

- 6. Train Staff on Business Continuity and Disaster Recovery Procedures:** CanTech should invest in regular training for its staff regarding their roles and responsibilities within the business continuity and disaster recovery plans. This is because every employee plays an important role in the execution of the business continuity plan. The proposed training should comprise drills and simulations. This will help to prepare them when/if they encounter real-world scenarios.
- 7. Establish Communication Plans and Protocols:** CanTech should deploy clear communication plans and protocols in order to keep stakeholders informed of different happenings. This is because effective communication is essential during and after a disaster. For this reason, the communication plans and protocols should entail:
 - the establishment of a crisis communication team,
 - the identification of key message points, and
 - the usage of multiple channels to communicate with employees, customers, and partners.
- 8. Partnership with External Support Services:** CanTech should consider serious partnerships with external support services including cloud service providers, cybersecurity firms, and disaster recovery specialists and in other areas where they lack internal expertise or resources.
- 9. Adoption of a Cloud-Based Continuity Strategy:** CanTech should leverage cloud computing for its scalability, flexibility, and cost-effectiveness in their disaster recovery settings. This is because implementing a cloud-based strategy can enhance CanTech's ability to resume operations as quickly as possible through the utilization of cloud infrastructure.
- 10. Integration of Regular Compliance Checks and Legal Reviews:** Since regulations and legal requirements are evolving, that is why it is important to incorporate regular reviews of compliance with laws that are crucial to data protection, cybersecurity, and business continuity. This will go a long way in ensuring that CanTech's disaster recovery and business continuity plans are not only operational but also legally compliant.
- 11. Optimize Supply Chain Resilience:** Critical suppliers and partners should be identified on the one hand, and contingency plans aimed at addressing potential supply chain disruptions should be developed on the other hand. Doing this might entail diversifying suppliers, establishing stockpiles of critical components, or agreeing on flexible contract terms to improve adaptability in crisis situations.
- 12. Advanced Analytics and AI for Predictive Insights should be Incorporated:** advanced analytics and artificial intelligence should be deployed to analyze trends, predict potential disruptions, and automate aspects of the disaster recovery process. With these technologies, early warnings can be provided, and more proactive management of risks can be enabled.
- 13. Establishment of a Dedicated Business Continuity Management Function:** A dedicated team or office responsible for business continuity management should be established. The essence of this team would be to oversee the development, implementation, and ongoing

management of business continuity strategies, ensuring a focused approach to organizational resilience.

- 14. Foster a Culture of Resilience:** In addition to formal training, a culture that values preparedness, adaptability, and resilience should be developed. To this end, employees at all levels should be encouraged to contribute ideas that would improve disaster recovery and business continuity practices, and those who play key roles in enhancing organizational resilience should be recognized.
- 15. Implementation of Energy Redundancy and Efficiency Measures:** CanTech data center should ensure that it has redundant power supplies like backup generators and uninterruptible power supply (UPS) systems. This will help in maintenance of operations during power outages. In addition, energy-efficient technologies should be adopted to help reduce operational costs and improve sustainability.

7 REFERENCES

1. “Roadmap for Success.” n.d. CanTech International.
<https://www.cantechonline.com/feature/33446/roadmap-for-success/>.
2. Peters, Silvana. 2023. “8 Phone Tree Templates for Small Business (+ Free Download).” Fit Small Business. March 16, 2023. <https://fitsmallbusiness.com/phone-tree-examples/>.
3. “How Do I Design and Initiate a Call Tree Procedure? | TechTarget.” n.d. Disaster Recovery. <https://www.techtarget.com/searchdisasterrecovery/answer/How-do-I-design-and-initiate-a-call-tree-procedure>.
4. Brush, Kate. 2022. “What Is a Disaster Recovery Plan (DRP) and How Do You Write One?” Edited by Paul Crocetti. TechTarget. 2022.
<https://www.techtarget.com/searchdisasterrecovery/definition/disaster-recovery-plan>.
5. Schmerler, Ben. 2021. “Types of Disasters to Consider for Your Disaster Recovery Plan.” Wwww.dpsolutions.com. March 24, 2021. <https://www.dpsolutions.com/blog/types-of-disasters>.
6. “Business Impact Analysis | Ready.gov.” 2023. Wwww.ready.gov. September 7, 2023.
<https://www.ready.gov/business/planning/impact-analysis>.
7. Kirvan, Paul. 2022. “How to Calculate Maximum Allowable Downtime.” SearchDisasterRecovery. March 28, 2022.
<https://www.techtarget.com/searchdisasterrecovery/tip/How-to-calculate-maximum-allowable-downtime>.
8. “Calculating the Real Cost of Downtime for Your Business | Ntirety.” n.d.
<https://www.ntirety.com/>. <https://www.ntirety.com/blog/calculating-the-real-cost-of-downtime-for-your-business/>.
9. Marget, Adam. 2021. “Downtime: Causes, Costs and How to Minimize It.” Unitrends. March 22, 2021. <https://www.unitrends.com/blog/downtime-causes-costs-and-how-to-minimize-it>.
10. kyndryl. 2016. “What Is a Disaster Recovery Plan? | Kyndryl.” Kyndryl.com. 2016.
<https://www.kyndryl.com/us/en/learn/disaster-recovery-plan>.
11. jincounterintuity. n.d. “What Is a Disaster Recovery Plan? 4 Examples.” Cloudian.
<https://cloudian.com/guides/disaster-recovery/4-disaster-recovery-plan-examples-and-10-essential-plan-items/amp/>.
12. “8 Must-Have Components of an Effective Disaster Recovery Plan.” 2020. Arcserve. August 13, 2020. <https://www.arcserve.com/blog/8-must-have-components-effective-disaster-recovery-plan>.
13. Tjader, Y., Han, S., & Vargas, L. (2020). Cloud computing in disaster recovery: strategy and readiness. *Journal of Business Continuity & Emergency Planning*, 13(4), 309-321.
14. Wallace, M., & Webber, L. (2017). *The disaster recovery handbook: A step-by-step plan to ensure business continuity and protect vital operations, facilities, and assets*. AMACOM.